The Replica Method II: Spin Glasses and the Phases of Learning

Physics of Learning • IIT Madras

Chapter 5

Chapter 4 developed the replica method on a single spin, where the $n \to 0$ limit was harmless and the overlap was a mere number. Here we restore the two missing ingredients—a thermodynamic limit and a *matrix-valued* overlap order parameter—using the Sherrington–Kirkpatrick (SK) model. We carry the calculation through in full: the disorder average, the birth of the overlap matrix q_{ab}, the construction of the saddle-point action, the replica-symmetric ansatz, the explicit $n \to 0$ algebra, and the self-consistency equation. We then show how a phase transition appears as a *bifurcation* of that equation, locating the spin-glass transition at $T_c = J$. Finally we map the machinery onto the Hopfield associative memory and read off its phase diagram, including the storage-capacity singularity at $\alpha_c \approx 0.138$.

The Sherrington–Kirkpatrick model

TAKE N SPINS $s_i = \pm 1$ with all-to-all random couplings:

$$H[\mathbf{s}] = -\sum_{i<j} J_{ij}\, s_i s_j, \qquad J_{ij} \sim \mathcal{N}\Big(0, \frac{J^2}{N}\Big) \text{ i.i.d.}$$

The $1/N$ scaling of the variance keeps the energy extensive.[1] The couplings J_{ij} are the quenched disorder, fixed for a given sample.

[1] Each spin interacts with $N-1$ others; for the total energy to be $O(N)$ rather than $O(N^2)$, the typical coupling must scale as $1/\sqrt{N}$, i.e. variance $\propto 1/N$.

WHY THIS IS THE MACHINE-LEARNING PROTOTYPE. In a Hopfield network or perceptron, the couplings are built from random training data $\{\boldsymbol{\xi}^\mu\}_{\mu=1}^P$—for instance the Hebbian rule $J_{ij} = \frac{1}{N}\sum_\mu \xi_i^\mu \xi_j^\mu$. The quenched disorder *is the dataset*. The spins are either neuron states (are we in a memory basin?) or the weights themselves (which configurations fit the data?). The disorder-averaged free energy therefore computes precisely the typical-case learning quantities. We treat generic SK first because the algebra is cleanest, then specialise to Hopfield in §.

Disorder average and the birth of the overlap

We want $f = -\frac{1}{\beta N}\overline{\ln Z}$, and apply replicas exactly as in Chapter 4:

$$\overline{Z^n} = \overline{\sum_{\{s^a\}} \exp\Big(\beta \sum_{a=1}^{n} \sum_{i<j} J_{ij}\, s_i^a s_j^a\Big)}.$$

The n replicas share the same couplings. Each J_{ij} appears linearly, so the disorder average is Gaussian; applying $\overline{e^{Jx}} = e^{\frac{1}{2}\mathrm{Var}(J)x^2}$ to each pair with $x = \beta \sum_a s_i^a s_j^a$ and $\mathrm{Var} = J^2/N$,

$$\overline{Z^n} = \sum_{\{s^a\}} \exp\left(\frac{\beta^2 J^2}{2N} \sum_{i<j} \Big(\sum_a s_i^a s_j^a \Big)^2 \right). \tag{1}$$

As before the disorder is gone, leaving a square. Expand it:

$$\Big(\sum_a s_i^a s_j^a \Big)^2 = \sum_{a,b} (s_i^a s_i^b)(s_j^a s_j^b),$$

and extend $\sum_{i<j} \to \frac{1}{2}\sum_{i,j}$ (the diagonal and double counting give $O(1)$ corrections, negligible against the $O(N)$ bulk):

$$\overline{Z^n} = \sum_{\{s^a\}} \exp\left(\frac{\beta^2 J^2}{4N} \sum_{a,b} \Big(\sum_i s_i^a s_i^b \Big)^2 \right). \tag{2}$$

THE OVERLAP IS BORN. The object $\sum_i s_i^a s_i^b$ that controls everything is, up to $1/N$, the *overlap* between replicas a and b:

$$q_{ab} \equiv \frac{1}{N} \sum_{i=1}^{N} s_i^a s_i^b.$$

In Chapter 4 this was the single number $s^a s^b$; now it is an $n \times n$ matrix, and it is the central object of the whole calculation. In terms of it,

$$\overline{Z^n} = \sum_{\{s^a\}} \exp\left(\frac{N\beta^2 J^2}{4} \sum_{a,b} q_{ab}^2 \right). \tag{3}$$

The exponent is now $O(N)$ and depends on the N microscopic spins *only through* the macroscopic matrix q_{ab}.

Decoupling the spins: the delta-function trick

Equation (3) cannot yet be summed over the spins, because the exponent contains $q_{ab}^2 = \big(\frac{1}{N}\sum_i s_i^a s_i^b\big)^2$, which couples *every site i to every site j* through the square. The strategy is to introduce q_{ab} as a new *independent* integration variable, so that the exponent becomes linear in the spins and the site sum factorises. The price is one constraint per replica pair, enforced by a Dirac delta.

A SINGLE SCALAR FIRST. Forget replicas for a moment and consider one macroscopic variable $Q = \frac{1}{N}\sum_i \phi_i$, where ϕ_i are some microscopic quantities. To trade the microscopic ϕ_i for the macroscopic Q, insert the identity[2]

[2] This is just $1 = \int dQ\, \delta(Q - \frac{1}{N}\sum_i \phi_i)$: the delta forces Q to equal the empirical mean. The factor of N inside, $\delta(NQ - \dots)$ versus $\delta(Q - \dots)$, only rescales the (irrelevant) normalisation.

$$1 = \int dQ\, \delta\Big(NQ - \sum_i \phi_i\Big). \tag{4}$$

Now write the delta in its *Fourier (integral) representation*,

$$\delta(x) = \int_{-i\infty}^{+i\infty} \frac{d\hat{Q}}{2\pi i}\, e^{-\hat{Q}\,x}, \tag{5}$$

where $\hat{Q}$ is a conjugate variable integrated along the imaginary axis.[3] Substituting (5) into (4) with $x = NQ - \sum_i \phi_i$ gives

$$1 = \int dQ\, d\hat{Q}\ \exp\Big(-\hat{Q}\big(NQ - \textstyle\sum_i \phi_i\big)\Big) = \int dQ\, d\hat{Q}\ \exp\Big(-N\hat{Q}\,Q + \hat{Q}\textstyle\sum_i \phi_i\Big). \tag{6}$$

[3] Equivalently $\delta(x) = \int \frac{dk}{2\pi} e^{-ikx}$ with $\hat{Q} = ik$. The imaginary-axis contour is what makes the later saddle-point well defined. We absorb constant prefactors like $2\pi i$ into the measure and stop writing them, since they contribute only $O(\ln N)$ to an $O(N)$ exponent.

The crucial feature: in the last form the microscopic variables ϕ_i now appear *linearly*, inside $\hat{Q}\sum_i \phi_i = \sum_i \hat{Q}\phi_i$ — a sum of independent single-site terms. The coupling has been moved into the new pair $(Q, \hat{Q})$.

Now the matrix version. We apply exactly this device to each replica pair. Here $\phi_i \to s_i^a s_i^b$ and $Q \to q_{ab}$, with a separate conjugate $\hat{q}_{ab}$ for every pair $a < b$ (the diagonal $q_{aa} = 1$ is fixed, since $(s_i^a)^2 = 1$, and needs no constraint). For each $a < b$ insert

$$1 = \int dq_{ab}\, d\tilde{q}_{ab}\ \exp\Big(-N\,\tilde{q}_{ab}\,q_{ab} + \tilde{q}_{ab}\textstyle\sum_i s_i^a s_i^b\Big), \tag{7}$$

where I temporarily call the conjugate $\tilde{q}_{ab}$. It is convenient to rescale it as $\tilde{q}_{ab} = \beta^2 J^2\, \hat{q}_{ab}$ so that later formulas are clean; this is just a change of integration variable. With that rescaling,

$$1 = \int dq_{ab}\, d\hat{q}_{ab}\ \exp\Big(-N\beta^2 J^2\, \hat{q}_{ab}\, q_{ab} + \beta^2 J^2\, \hat{q}_{ab}\textstyle\sum_i s_i^a s_i^b\Big). \tag{8}$$

Insert into $\overline{Z^n}$. Multiply Eq. (3) by one factor of (8) for each pair $a < b$. The exponent now splits into three groups of terms:

$$\overline{Z^n} = \int \prod_{a<b} dq_{ab}\, d\hat{q}_{ab}\ \exp\Big(\underbrace{\tfrac{N\beta^2 J^2}{4}\textstyle\sum_{a,b} q_{ab}^2}_{\text{(i) from (3)}} - \underbrace{N\beta^2 J^2 \textstyle\sum_{a<b}\hat{q}_{ab} q_{ab}}_{\text{(ii) constraint, } q\text{-part}}\Big)\ \underbrace{\sum_{\{s^a\}} \exp\Big(\beta^2 J^2 \textstyle\sum_{a<b}\hat{q}_{ab}\sum_i s_i^a s_i^b\Big)}_{\text{(iii) constraint, spin-part}}. \tag{9}$$

Term (i) replaced the empirical q_{ab} by the independent variable, which is legitimate *because* the delta enforces their equality. Terms (i) and (ii) depend only on $\{q, \hat{q}\}$ and pull out of the spin sum. All the spin dependence is now in (iii), and—this is the payoff—it is *linear* in the bilinears $s_i^a s_i^b$.

Why the spin part factorises over sites. Look at the exponent of (iii) and exchange the order of summation, pulling the site sum outward:

$$\beta^2 J^2 \sum_{a<b} \hat{q}_{ab} \sum_i s_i^a s_i^b \;=\; \sum_{i=1}^{N} \underbrace{\Big(\beta^2 J^2 \sum_{a<b} \hat{q}_{ab}\, s_i^a s_i^b\Big)}_{\text{depends on site } i \text{ only}}.$$

The total exponent is a *sum over sites* of a term involving only the n replica-spins at that one site. The exponential of a sum is a product of exponentials, and the spin sum $\sum_{\{s^a\}} = \prod_i \sum_{\{s_i^a\}}$ also factorises over sites, so

$$\sum_{\{s^a\}} \exp\Big(\sum_i \beta^2 J^2 \sum_{a<b} \hat{q}_{ab} s_i^a s_i^b\Big) = \prod_{i=1}^{N} \sum_{\{s_i^a\}} \exp\Big(\beta^2 J^2 \sum_{a<b} \hat{q}_{ab} s_i^a s_i^b\Big).$$

Every factor in the product is *identical*—the site label i is a dummy, since each site sees the same $\hat{q}_{ab}$. Calling the common single-site sum

$$\zeta[\hat{q}] = \mathrm{Tr}_{\{s^a\}} \exp\Big(\beta^2 J^2 \sum_{a<b} \hat{q}_{ab}\, s^a s^b\Big), \qquad \mathrm{Tr}_{\{s^a\}} \equiv \sum_{s^1=\pm 1} \cdots \sum_{s^n=\pm 1}, \tag{10}$$

the product is simply $\zeta[\hat{q}]^N$. (Here s^a denotes the n replica spins *at one site*; the 2^n terms of the trace are all the configurations of those n spins.)

THE ACTION. Writing $\zeta^N = e^{N \ln \zeta}$ and collecting (i), (ii), (iii), we obtain $\overline{Z^n} = \int \prod_{a<b} dq_{ab}\, d\hat{q}_{ab}\, e^{N\Phi[q,\hat{q}]}$ with

$$\boxed{\Phi[q,\hat{q}] = \frac{\beta^2 J^2}{4} \sum_{a,b} q_{ab}^2 - \beta^2 J^2 \sum_{a<b} \hat{q}_{ab}\, q_{ab} + \ln \zeta[\hat{q}].}$$

Three pieces: the bare q^2 energy (i), the Legendre coupling between q and its conjugate (ii), and the single-site term $\ln \zeta$ (iii). Every term is $O(N)$ in the exponent, which is what makes the saddle-point evaluation of the next section exact as $N \to \infty$.

The saddle point

Because every term in the exponent is $O(N)$, the integral is dominated as $N \to \infty$ by its saddle point (Laplace's method): $\overline{Z^n} \approx e^{N\Phi[q^\star,\hat{q}^\star]}$, with Φ extremised. *This is the qualitative break from Chapter 4*: there was no N there, hence no saddle; here the saddle is the physics.

EXTREMISE OVER q_{ab} (off-diagonal):

$$\frac{\partial \Phi}{\partial q_{ab}} = 0 \;\Rightarrow\; \beta^2 J^2 q_{ab} - \beta^2 J^2 \hat{q}_{ab} = 0 \;\Rightarrow\; \hat{q}_{ab} = q_{ab}.$$

The conjugate field simply equals the overlap at the saddle.

EXTREMISE OVER $\hat{q}_{ab}$:

$$\frac{\partial \Phi}{\partial \hat{q}_{ab}} = 0 \;\Rightarrow\; q_{ab} = \frac{1}{\zeta}\,\mathrm{Tr}\left[s^a s^b\, e^{\beta^2 J^2 \sum_{c<d} \hat{q}_{cd} s^c s^d}\right] = \langle s^a s^b \rangle_{\mathrm{eff}}.$$

This is the *self-consistency equation*: the overlap equals the thermal average of $s^a s^b$ in an effective single-site problem whose couplings are the overlaps themselves. The loop is closed—but to solve it, and to take $n \to 0$, we must assume a structure for the matrix.

The replica-symmetric ansatz

THE SIMPLEST GUESS is that all replicas are equivalent (*replica symmetry*, RS):

$$q_{ab} = q \ (a \neq b), \qquad q_{aa} = 1, \qquad \hat{q}_{ab} = q \ (a \neq b).$$

We now evaluate each piece of Φ as an explicit function of n and q.

PIECE 1 — THE q^2 TERM. The sum $\sum_{a,b} q_{ab}^2$ runs over all n^2 entries: n diagonal entries equal to 1, and $n(n-1)$ off-diagonal entries equal to q^2. Thus

$$\frac{\beta^2 J^2}{4} \sum_{a,b} q_{ab}^2 = \frac{\beta^2 J^2}{4} \left[n + n(n-1) q^2 \right].$$

PIECE 2 — THE COUPLING TERM. The sum over $a < b$ has $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs, each contributing $\hat{q}_{ab} q_{ab} = q^2$:

$$-\beta^2 J^2 \sum_{a<b} \hat{q}_{ab} q_{ab} = -\beta^2 J^2 \, \frac{n(n-1)}{2} \, q^2.$$

PIECE 3 — THE SINGLE-SITE TRACE. We must evaluate

$$\zeta = \operatorname{Tr} \exp\Big(\beta^2 J^2 q \sum_{a<b} s^a s^b \Big).$$

Use $\sum_{a<b} s^a s^b = \frac{1}{2}\left[(\sum_a s^a)^2 - n \right]$ (since $(s^a)^2 = 1$):

$$\zeta = e^{-\beta^2 J^2 q\, n/2} \; \operatorname{Tr} \exp\Big(\tfrac{\beta^2 J^2 q}{2} \big(\textstyle\sum_a s^a \big)^2 \Big).$$

The square couples replicas; *Hubbard–Stratonovich* decouples them, exactly as in Chapter 4:[4]

$$\zeta = e^{-\beta^2 J^2 q\, n/2} \int \mathcal{D}z \left[\operatorname{Tr}_s \, e^{\beta J \sqrt{q}\, z\, s} \right]^n = e^{-\beta^2 J^2 q\, n/2} \int \mathcal{D}z \, \left[2 \cosh(\beta J \sqrt{q}\, z) \right]^n.$$

[4] $e^{\frac{1}{2} a m^2} = \int \mathcal{D}z \, e^{\sqrt{a}\, z m}$ with $a = \beta^2 J^2 q$, so $\sqrt{a} = \beta J \sqrt{q}$.

The same integral structure as the toy model, but now the effective field strength $\beta J \sqrt{q}$ is self-generated by the overlap rather than externally fixed.

Taking $n \to 0$ explicitly

We need $f = -\frac{1}{\beta} \lim_{n \to 0} \frac{1}{n} \Phi$. Take each piece divided by n, using $n(n-1)/n = n - 1 \to -1$.

THE POLYNOMIAL PIECES:

$$\frac{1}{n}\left[\tfrac{\beta^2J^2}{4}\big(n+n(n-1)q^2\big)-\tfrac{\beta^2J^2}{2}n(n-1)q^2\right]\xrightarrow{n\to 0}\tfrac{\beta^2J^2}{4}(1-q^2)+\tfrac{\beta^2J^2}{2}q^2$$
$$=\tfrac{\beta^2J^2}{4}\big(1+q^2\big).$$

THE $\ln\zeta$ PIECE. Write $\zeta=e^{-\beta^2J^2q\,n/2}I_n$ with $I_n=\int\mathcal{D}z\,[2\cosh(\beta J\sqrt{q}\,z)]^n$. Since $I_0=\int\mathcal{D}z=1$, we have $\ln I_n=n\,\partial_n I_n|_0+O(n^2)$ with $\partial_n[2\cosh]^n|_0=\ln(2\cosh)$. Hence

$$\frac{1}{n}\ln\zeta\xrightarrow{n\to 0}-\frac{\beta^2J^2q}{2}+\int\mathcal{D}z\,\ln\big[2\cosh(\beta J\sqrt{q}\,z)\big].$$

COLLECTING (and using $\frac{\beta^2J^2}{4}(1+q^2)-\frac{\beta^2J^2}{2}q=\frac{\beta^2J^2}{4}(1-q)^2$), the replica-symmetric free energy is

$$\boxed{-\beta f=\frac{\beta^2J^2}{4}(1-q)^2+\int\mathcal{D}z\,\ln\big[2\cosh(\beta J\sqrt{q}\,z)\big].}$$

Consistency check. Formally setting $q=0$ collapses this to $\frac{\beta^2J^2}{4}+\ln 2$—the Chapter 4 answer with $\sigma\to J$. The toy model was the $q=0$ slice of the spin glass.

The order-parameter equation

Extremise the free energy over q. Differentiating the integral and integrating by parts in z[5] gives the clean fixed-point equation

$$\boxed{q=\int\mathcal{D}z\,\tanh^2\big(\beta J\sqrt{q}\,z\big).}$$

[5] Using $\frac{d}{dq}\ln 2\cosh(\beta J\sqrt{q}z)=\tanh(\beta J\sqrt{q}z)\cdot\frac{\beta Jz}{2\sqrt{q}}$, then $\int\mathcal{D}z\,z\,g(z)=\int\mathcal{D}z\,g'(z)$ (Gaussian integration by parts), and $\frac{d}{dz}\tanh=\beta J\sqrt{q}\,\mathrm{sech}^2$. The $\frac{\beta^2J^2}{2}$ terms cancel, leaving the result.

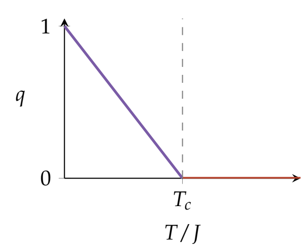
Compare the toy-model magnetisation $\overline{\langle s\rangle^2}=\overline{\tanh^2(\beta h)}$: identical in form, but now q appears on *both* sides. The overlap is self-generated—each spin feels an effective random field $\beta J\sqrt{q}\,z$ created collectively by all the others through q itself.[6]

[6] This is the *cavity-field* picture: remove one spin, and the field it would feel is a Gaussian of variance J^2q produced by the rest of the system.

Where the singularity is: the spin-glass transition

We now locate the non-analyticity. Set $J=1$ and study the fixed-point equation as $\beta=1/T$ varies.

$q=0$ IS ALWAYS A SOLUTION (the paramagnet): if $q=0$ the tanh argument vanishes and $\tanh^2(0)=0$, consistent. The transition is the point where a *nonzero* branch splits off.



The order parameter q bifurcates from zero at $T_c=J$. For $T>T_c$: paramagnet ($q=0$). For $T<T_c$: spin glass ($q>0$).

EXPAND FOR SMALL q. For small argument $\tanh^2(x) \approx x^2 = \beta^2 q z^2$, so

$$q = \int \mathcal{D}z\, \beta^2 q\, z^2 = \beta^2 q \underbrace{\int \mathcal{D}z\, z^2}_{=1} = \beta^2 q.$$

Thus $q(1-\beta^2) = 0$: a nonzero solution becomes possible exactly when the coefficient reaches unity,

$$\boxed{\beta_c^2 = 1 \quad \Longrightarrow \quad T_c = J.}$$

This is the singularity. For $T > T_c$ only $q = 0$ exists (paramagnetic phase: disordered, no frozen structure). For $T < T_c$ a stable $q > 0$ branch appears (spin-glass phase: spins freeze into a random but rigid pattern). The free energy $f(T)$ is analytic on each side but has a non-analytic kink at T_c—its second derivative is discontinuous. That discontinuity *is* the phase transition, and it arose purely from the bifurcation of the saddle-point equation.

THE GENERAL MECHANISM. A phase transition appears in this formalism wherever the *number or stability of solutions* to the saddle-point equation changes. A new branch appearing (or an existing one vanishing) is a singularity of $\overline{\ln Z}$. This is the precise content of the slogan "non-analyticities of the free energy are phase transitions."

A caveat: when replica symmetry fails

The RS ansatz is not always correct. Computing the entropy from the RS free energy, one finds it goes *negative* at low temperature—impossible for a discrete spin system.[7] The implication for the $n \to 0$ trick is sharp: the continuation is *not* innocent. The structure one assumes for the matrix as $n \to 0$ determines the answer, and the correct structure (RS, or one of the RSB schemes) is a nontrivial property of the problem. The full Parisi treatment is beyond these notes; what matters here is that the *location of a transition*—a bifurcation of the order-parameter equation—is already visible at RS level.

[7] This is the de Almeida–Thouless instability: below a line in the (T, h) plane the RS saddle is no longer a maximum in replica space. The cure is Parisi's *replica symmetry breaking*, in which q_{ab} is not uniform but hierarchically structured, described by a function $q(x)$, $x \in [0,1]$, and the order parameter becomes a whole distribution of overlaps $P(q)$.

The phases of learning: Hopfield memory

NOW MAKE THE DISORDER DATA. Store $P = \alpha N$ random patterns $\{\boldsymbol{\xi}^\mu\}$, $\xi_i^\mu = \pm 1$, via the Hebbian couplings

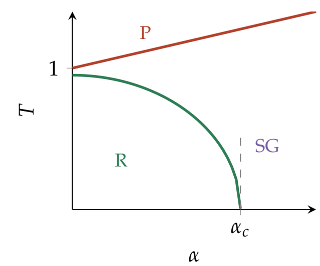
$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu.$$

The quenched disorder is now the stored dataset. The replica calculation proceeds as above, but with *two* kinds of order parameter:

- the *retrieval overlap* $m = \frac{1}{N}\sum_i \langle s_i \rangle \xi_i^1$ with the pattern we are trying to recall;
- the *spin-glass overlap* $q = \frac{1}{N}\sum_i \langle s_i \rangle^2$ generated by the other $P - 1$ patterns acting as quenched noise.

The interference from the non-retrieved patterns plays exactly the role of the random field in the toy model—a self-consistent Gaussian noise of variance set by α and q.

THE PHASE DIAGRAM in the (α, T) plane has three regions, each a different solution branch of the saddle-point equations.



Hopfield phases: Paramagnetic, Retrieval, Spin-Glass. Retrieval ends at $\alpha_c \approx 0.138$.

Retrieval (R) — low α, low T: a solution with $m \approx 1$ exists; the network recalls the stored memory. The free energy has a minimum at large overlap with a pattern.

Spin glass (SG) — high α: only $m = 0$, $q > 0$ solutions survive; the network freezes into spurious mixtures and memories are destroyed.

Paramagnet (P) — high T: $m = q = 0$; thermal noise melts all structure.

THE CAPACITY SINGULARITY. The retrieval/spin-glass boundary at low temperature sits at the storage capacity

$$\boxed{\alpha_c \approx 0.138.}$$

This number is precisely the value of α at which the $m \neq 0$ retrieval branch of the saddle-point equations *ceases to exist*—the same bifurcation mechanism as the SK transition in §, now in the data-load parameter α rather than temperature. Beyond α_c the memory is overloaded: adding more patterns destroys the ability to recall any of them. The crossover that was perfectly smooth for one neuron in Chapter 4 has become a sharp, computable threshold.

The throughline

The single spin of Chapter 4 taught the *mechanism*: average the log via replicas; the shared disorder couples them; an overlap is born. This chapter added the two ingredients that make the method powerful and genuinely hard:

1. a *thermodynamic limit*, turning the calculation into a saddle point over the overlap;

2. a *matrix order parameter* q_{ab}, whose $n \to 0$ structure—replica-symmetric or broken—encodes the real physics.

And the payoff is concrete: a phase transition is a bifurcation of the order-parameter equation, and quantities such as the Hopfield capacity $\alpha_c \approx 0.138$ are read directly off where a solution branch of $\overline{\ln Z}$ appears or vanishes. The free energy is, in the end, the generating function for everything a learning system typically does.